Airbnb Price Prediction

Dataset-driven regression experiment

1. Objective

Predict the supplied dataset's log-transformed Airbnb listing price from listing characteristics, location, reviews, amenities and textual information.

2. Dataset

The experiment uses 74,111 listings from NYC, LA, SF, DC, Chicago and Boston. The target is the dataset's existing **log_price** field. The data are split 80:20 using city-stratified sampling.

3. Method

Numeric variables are median-imputed and standardized. Categorical variables are one-hot encoded. Amenities and listing text are represented with TF-IDF. All preprocessing is fitted only on the training partition to prevent test leakage. Ridge regression, Random Forest and XGBoost are evaluated on the held-out test set.

4. Results

Model	MAE	RMSE	R^2	Time (s)
ridge	0.2857	0.3967	0.6992	4.21

5. Conclusion

The reported metrics are produced from the supplied dataset and the code in this repository. No performance numbers from an external publication are used as experimental results.